Putting it all together

$\left\{ \text{Conjugacy classes of } A \in M_{n\times n}(k) \right\} \longleftrightarrow \left\{ \text{Classifying isom. classes of } k\text{-pairs } (k^n, T_A) \right\}$

$\updownarrow$ (left) $\qquad\qquad \updownarrow$ (right)

$\left\{ \text{Classifying } k[x]\text{-modules } M \text{ up to isom. where the underlying } k\text{-v. space is } n\text{-dimensional} \right\} \longleftrightarrow \left\{ \text{Classifying isom. classes of } k\text{-pairs } (V, T),\ \dim_k V = n \right\}$

T is telling us how to multiply by x

Observation: (1) $k[x]$ is a PID

(2) M is a $k[x]$-module

$\dim_k M = n < \infty$

Then M is a finitely generated $k[x]$-module

Our classification problem is now about classifying f.g. $k[x]$-modules

Theorem (FT for f.g. $k[x]$-modules) invariant factor formulation

M: f.g. $k[x]$-module

Then $\exists!$ $f_1(x) \mid f_2(x) \mid \dots \mid f_r(x)$ with leading coeff. 1 & $\deg \geq 1$

and $\exists!$ $m \geq 0$

s.t. $M \simeq \overbrace{k[x]/(f_1(x)) \times \dots \times k[x]/(f_r(x))}^{\text{fin. dim } k\text{-v. spaces}}$

k[x]-module isom.

$\times \underbrace{k[x]^m}_{\substack{\text{infinite dim} \\ k\text{-v. space} \\ \text{if } m \geq 1.}}$

Observation ③

$$M_{tors} \simeq \overbrace{k[x]/(f_1(x)) \times \cdots \times k[x]/(f_r(x))}^{\dim = \sum_{i=1}^{r} \deg(f_i(x))}$$

$$\boxed{M_{tf} \simeq k[x]^m}$$

← will not show up for k[x]-mod arising from fin. dim'l k-pairs

Observation ④

If M is obtained from a k-pair (V, T)

then $M = M_{tors} \simeq k[x]/(f_1(x)) \times \cdots \times k[x]/(f_r(x))$

and $\sum_{i=1}^{r} \deg(f_i(x)) = \dim_k V$

Unwinding the equivalences

$$A \in M_{n\times n}(k)$$

$$\Big\downarrow$$

$$(k^n, T_A) \rightsquigarrow k[x]\text{-module } M_A$$

$M_A = k^n$ as a k-v.space

with $x \cdot v = Av$

$$\Big\downarrow \text{FT}$$

$$M_A \simeq k[x]/(f_1(x)) \times \cdots \times k[x]/(f_r(x))$$

$$f_1(x) \mid \cdots \mid f_r(x)$$

uniquely determined by A

$$\Longrightarrow (k^n, T_A) \qquad \Big(k[x]/(f_1(x)), \underset{\substack{\uparrow \\ \text{mult. by } x}}{T}\Big)$$

$\Vert$ defn $\qquad \Vert$

$$V_A \simeq V_{f_1} \times \cdots \times V_{f_r}$$

$$f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$$

$$M_f = \begin{bmatrix} 0 & 0 & & 0 & -a_0 \\ 1 & 0 & & \vdots & -a_1 \\ 0 & 1 & \ddots & & \vdots \\ \vdots & 0 & \ddots & 0 & \vdots \\ 0 & 0 & & 1 & -a_{n-1} \end{bmatrix} \simeq V_{M_{f_1}} \times \cdots \times V_{M_{f_r}} \simeq V_{\begin{bmatrix} M_{f_1} & & 0 \\ & M_{f_2} & \\ & & \ddots \\ 0 & & & M_{f_r} \end{bmatrix}}$$

$\iff$ A is k-conjugate

to $\begin{bmatrix} M_{f_1} & & & 0 \\ & M_{f_2} & & \\ & & \ddots & \\ 0 & & & M_{f_r} \end{bmatrix}$ $\leftarrow$ Rational canonical form

Observation 5 $A, B \in M_{n\times n}(k)$

A & B are k-conjugate

$\iff$ They have the same RCF

Def^n $f_1(x) \mid \dots \mid \boxed{f_r(x)}$: invariant factors of A

minimal polynomial $\min_A(x)$

$f_1(x) f_2(x) \cdots f_r(x)$

characteristic polynomial $ch_A(x)$

Example:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$k^2 = M_A$$

$$x \cdot v = x \cdot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a+b \\ b \end{bmatrix}$$

$$k[x]/(x-\lambda) \times k[x]/(x-\lambda)$$

$$\|$$

$$M_A \simeq \begin{cases} k[x]/(f(x)) \times k[x]/(f(x)) & \deg f(x) = 1 \\ \quad \text{or} & \\ k[x]/(f(x)) & \deg f(x) = 2 \end{cases}$$

$$f(x) = x^2 + a_1 x + a_0$$

$$\simeq \begin{cases} M_{\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}} \\ M_{\begin{bmatrix} 0 & -a_0 \\ 1 & -a_1 \end{bmatrix}} \end{cases}$$

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \rightsquigarrow \begin{array}{c} min_A(x) \\ \| \\ ch_A(x) = (x-1)^2 \end{array}$$